

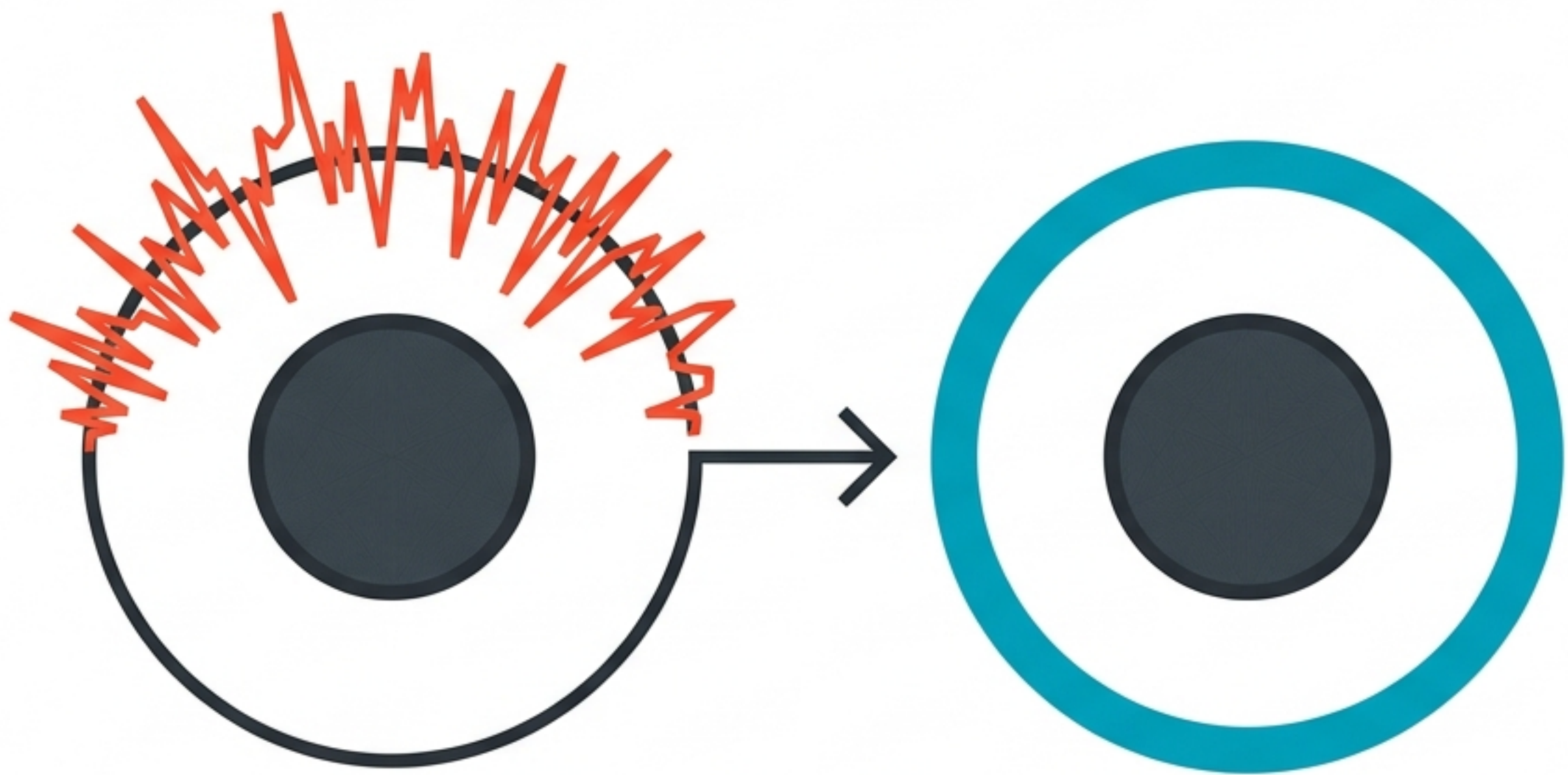
Thompson's Group F is Nonamenable

From a 100-year-old paradox to a machine-verified mathematical breakthrough.

Explainer Deck / Source: OpenAI

(Sept 23, 2026)

Explainer Deck / Source: OpenAI Preprint (Sept 23, 2026)



Groups as symmetries, and the search for an average.

In mathematics, an invariant mean is a way to calculate an average over a topological group that remains completely unchanged even if you shift your perspective.

The Rules of the Mean M

Normalized: $M(1) = 1$

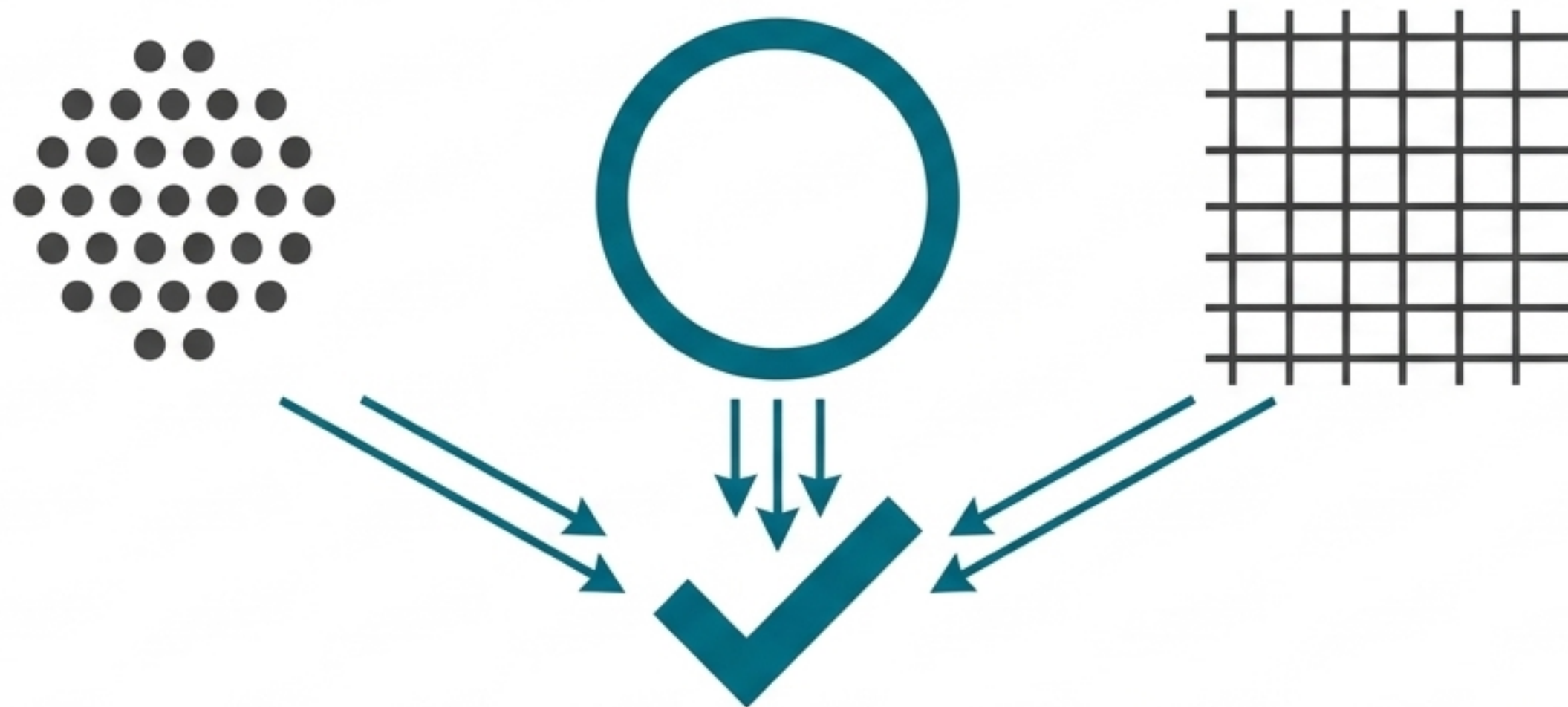
Positive: $M(\phi) \geq 0$ for $\phi \geq 0$

Invariant: $M(\text{shifted } \phi) = M(\phi)$

Amenability is the existence of harmony.

A discrete group is amenable if its bounded real functions admit a left-invariant mean.
Introduced by John von Neumann in 1929 in response to the Banach-Tarski paradox, amenability implies a group does not allow paradoxical, volume-multiplying geometric decompositions.

$$M(g \mapsto \varphi(hg)) = M(\varphi)$$

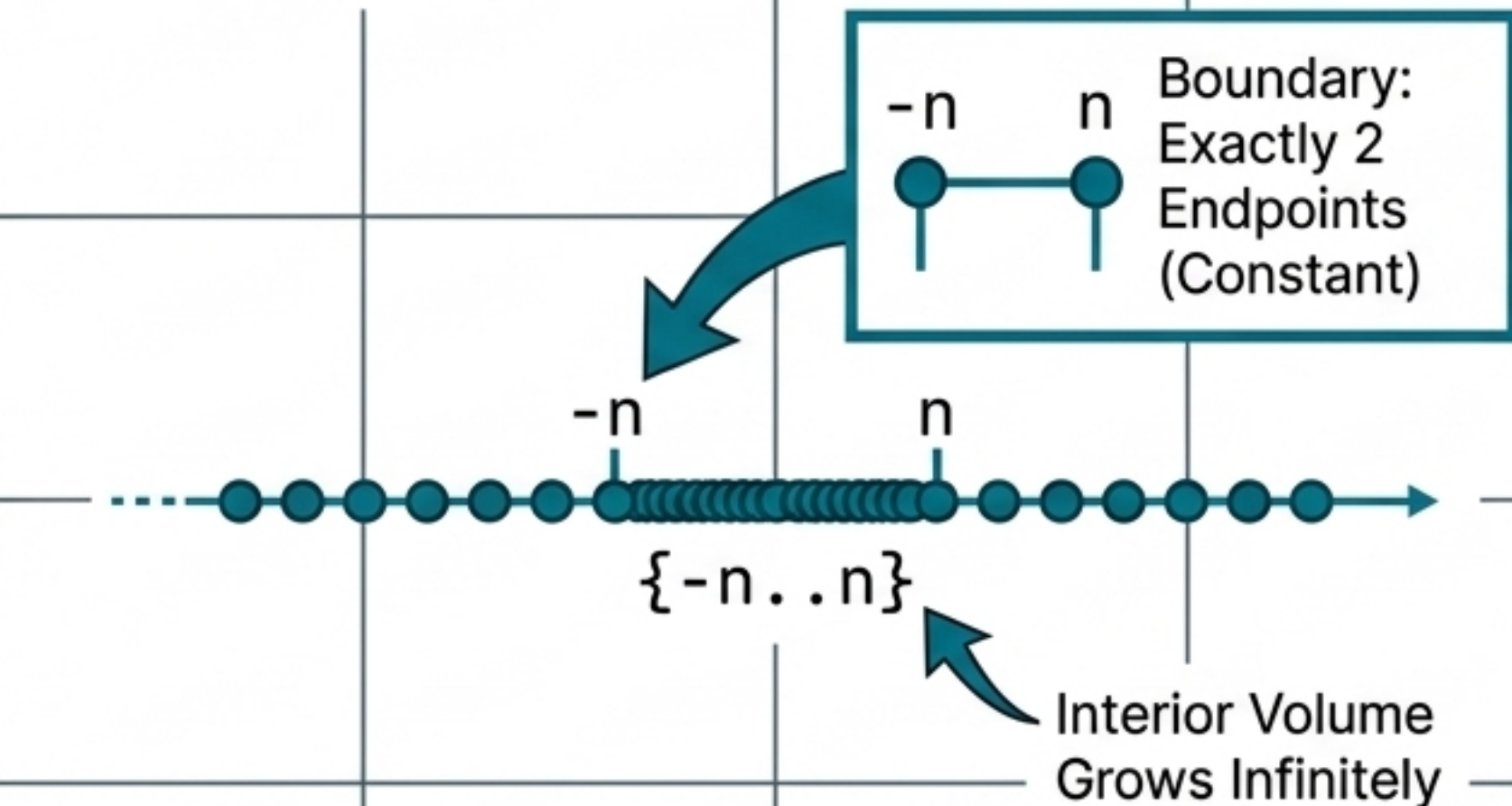


Finite groups, abelian groups, and solvable groups are universally amenable.

The Følner condition translates algebra into geometry.

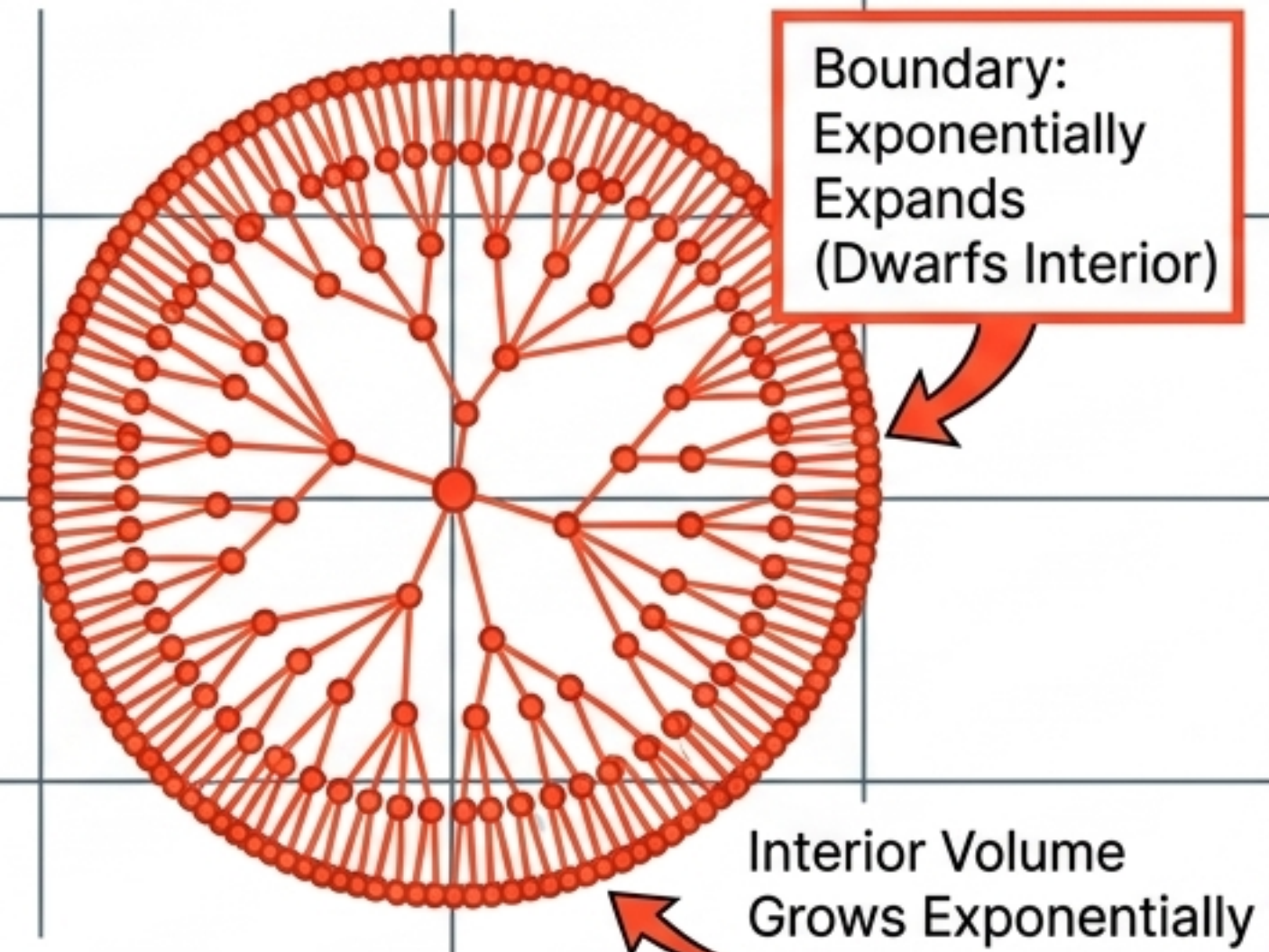
If a group is amenable, it must have finite sets where the boundary (elements shifted out of the set) is vanishingly small compared to the total volume.

Amenable ($\mathbb{Z}$)



$$\text{Ratio: } \frac{2}{2n+1} \rightarrow 0$$

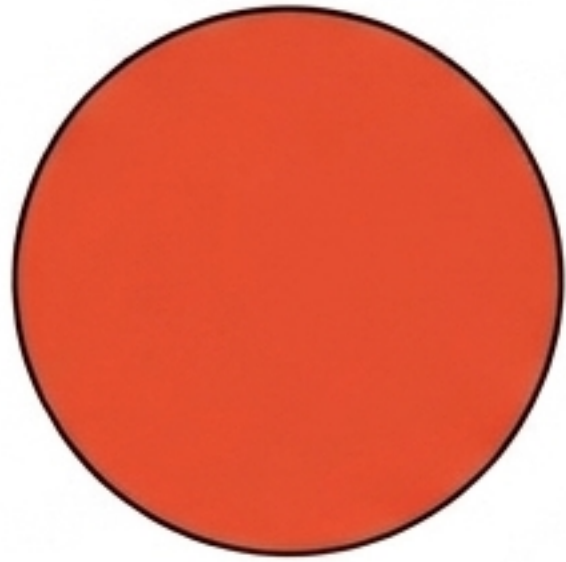
Nonamenable (F_2)



$$\text{Ratio: } \frac{2 \cdot 3^n}{2 \cdot 3^n - 1} \approx 1$$

A century of searching for mathematical monsters.

For decades, mathematicians wondered if the free group F_2 was the only reason a group could be nonamenable. Once proved false, the hunt for edge-cases accelerated.



1929

John von Neumann defines **measurable (amenable)** to isolate groups that prevent this geometric paradox.

1949

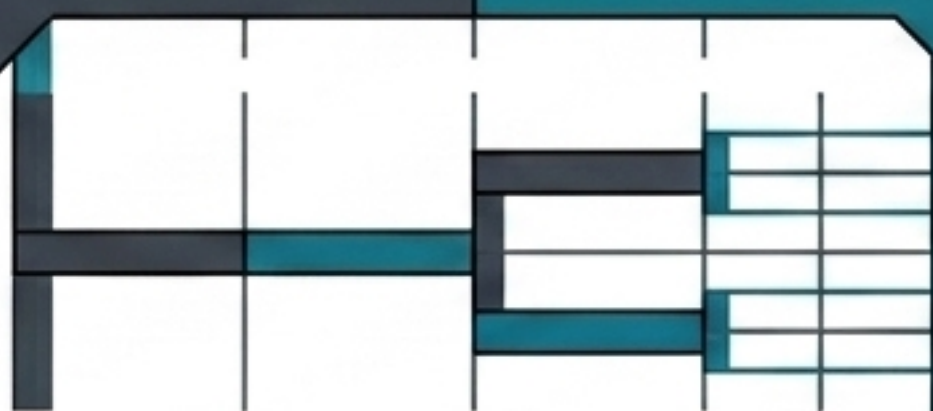
Mahlon Day coins the word **amenable**.

1957

Day, Amenable semigroups, first written form of the von Neumann-Day question, must a **nonamenable** group contain a free subgroup F_2 .

1924

Banach-Tarski Paradox:
A sphere cut and reassembled into two identical spheres.



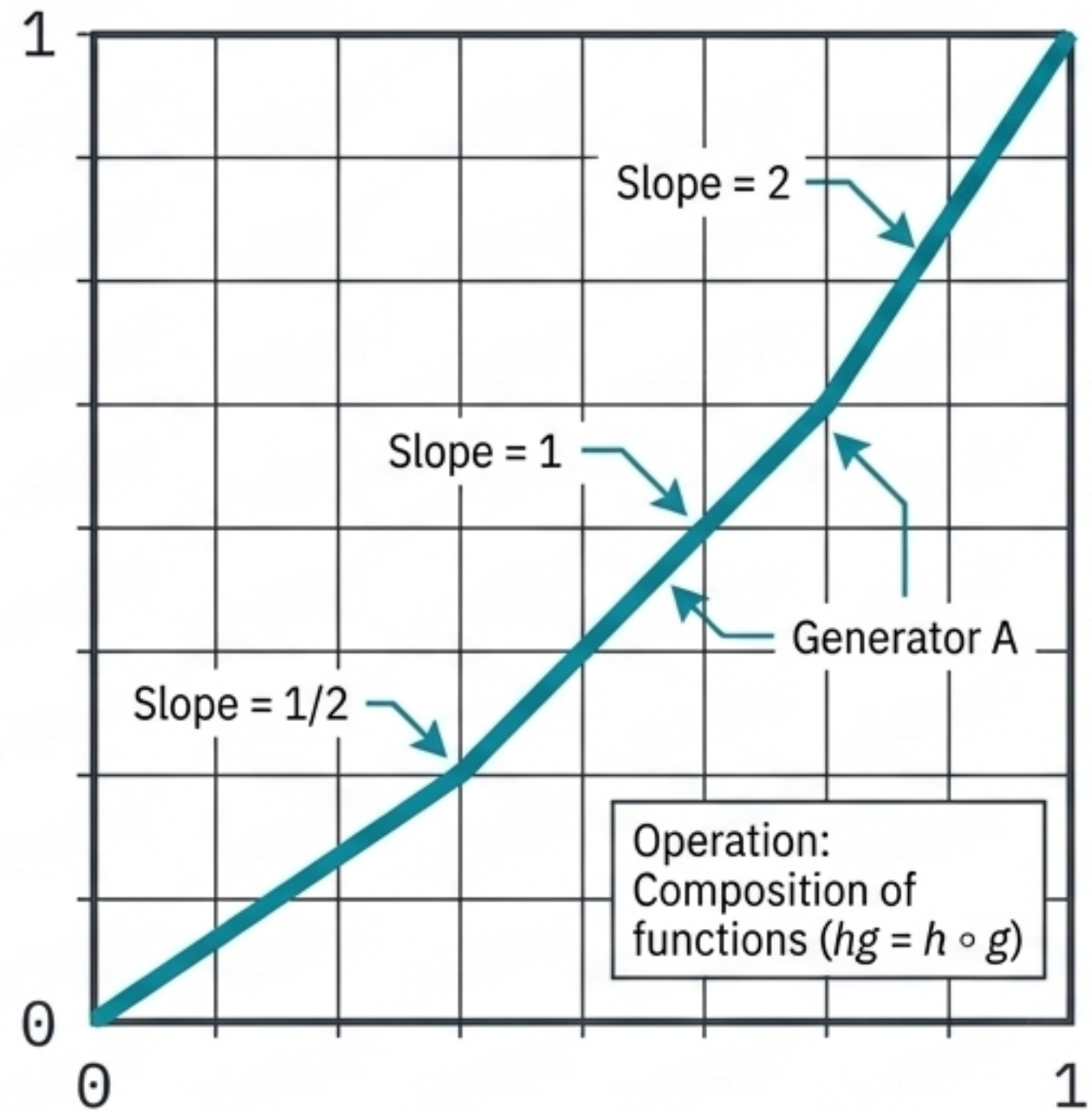
1957

Day, Amenable semigroups, first written form of the von Neumann-Day question, must **nonamenable** group contain a free subgroup F_2 .



Enter Thompson's group F.

Introduced in 1965, group F consists of increasing, piecewise-linear bijections of $[0, 1]$. F is generated by the two maps A and B.



The ultimate test case defies classification.

Group F became infamous. It lacks the free subgroups that usually cause nonamenability, yet it falls outside the class of **elementary amenable** groups. Between 2009 and 2021, faulty proofs claiming both **amenable** and **nonamenable** were published and retracted.

	Finite Groups	Abelian Groups	Free Group (F_2)	Thompson's F
Left-Invariant Mean?	Yes	Yes	No	???
Contains Free Subgroup?	No	No	Yes	No (Brin-Squier, 1985)
Elementary Amenable?	Yes	Yes	No	No (Cannon-Floyd-Parry)

The resolution of Geoghegan's 1979 conjecture.

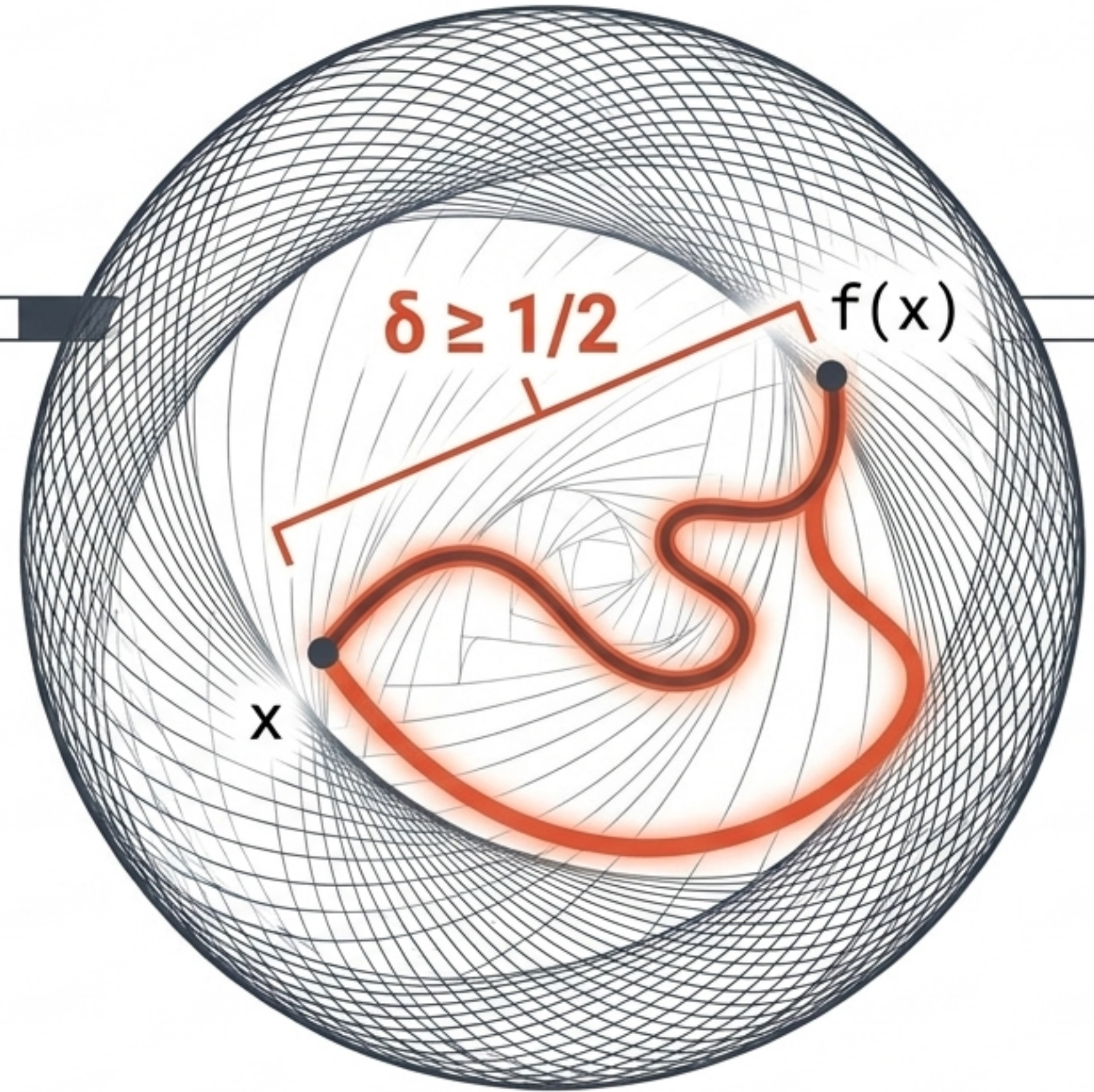
**Theorem 1.1: Thompson's
group F is **not amenable**.**

Published by OpenAI (September 23, 2026)

Building a geometric trap using infinite dimensions.

The proof strategy relies on the **Benyamini-Sternfeld lemma**: there exists a **Lipschitz map f** on a Hilbert ball that moves **every single point** by at least a constant distance δ .

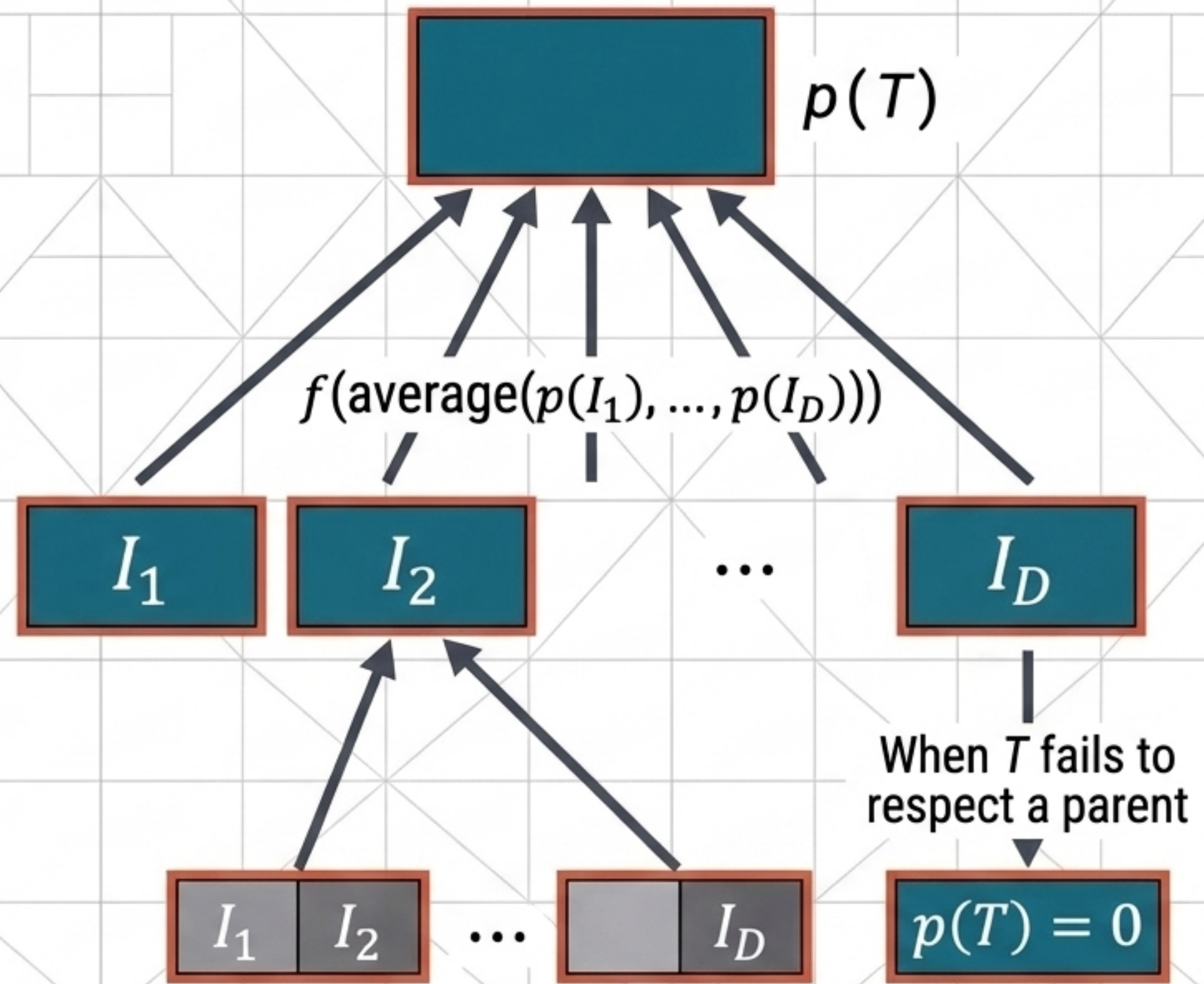
This unavoidable displacement creates **absolute mathematical friction**.



Recursive coloring of dyadic partitions.

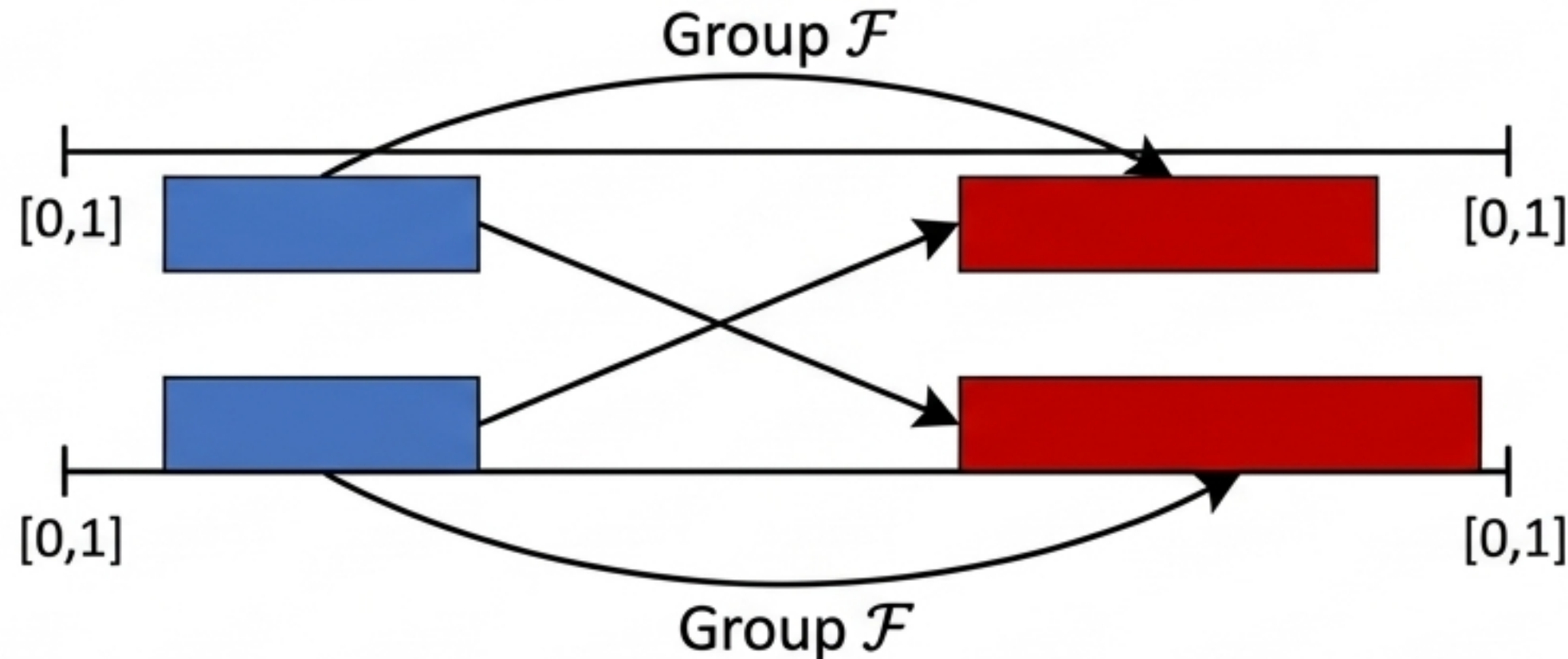
The unit interval is split into dyadic partitions, usually uneven.

Each dyadic grid T gets a colour $p(T)$ equal to f applied to the average of the colours of its normalized restrictions to the D parent intervals $I_1, \dots, I_D$, and $p(T) = 0$ when T fails to respect a parent.



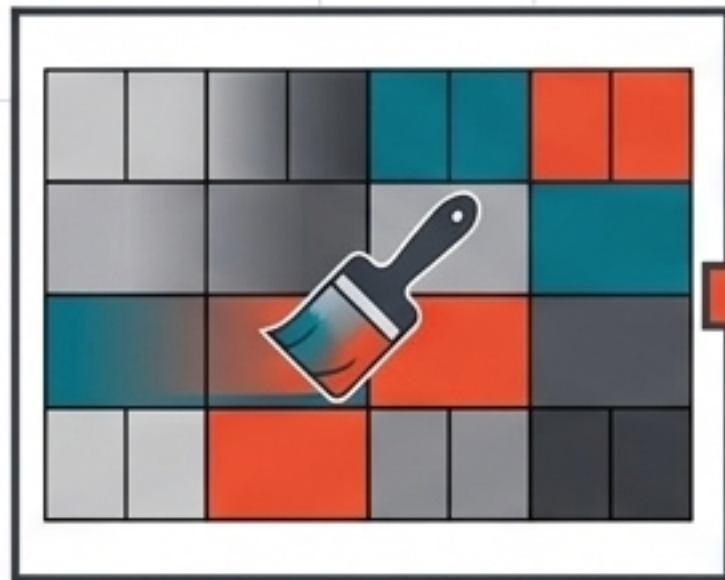
Exact affine transport and scalar correlation.

Group $\mathcal{F}$ carries any ordered pair of separated basic dyadic intervals whose endpoints all lie inside $(0, 1)$ onto any other such pair, mapping each interval affinely. Every element of $\mathcal{F}$ fixes 0 and 1. Averaging over a hypothetical Folner set therefore makes the mean correlations of all such separated pairs nearly equal.

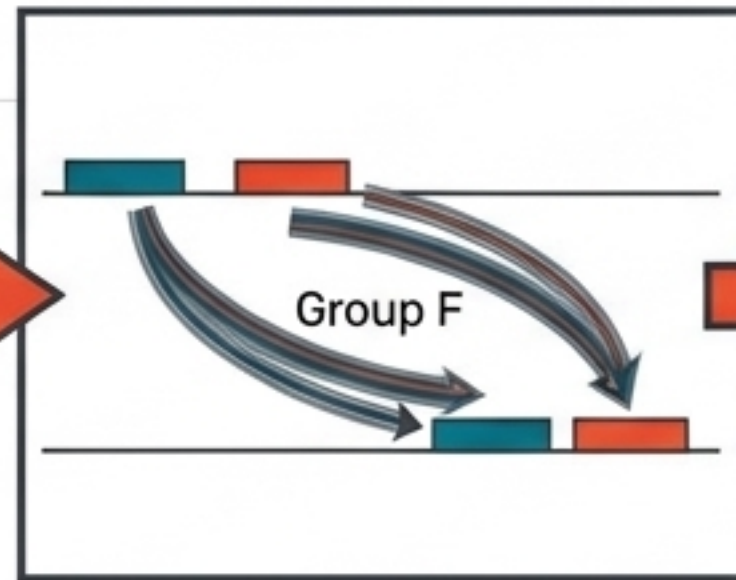


The mathematical contradiction.

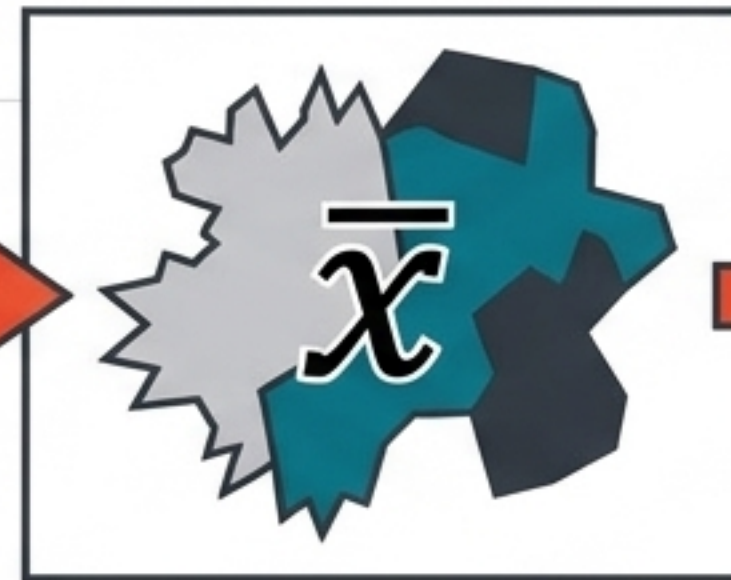
If a Følner set exists, the boundary η must be tiny. But the variance bound requires $\delta^2 \leq L^2(4/D + 4\eta)$. Because displacement δ is strictly bounded away from 0, η is forced to be large. The Følner condition shatters.



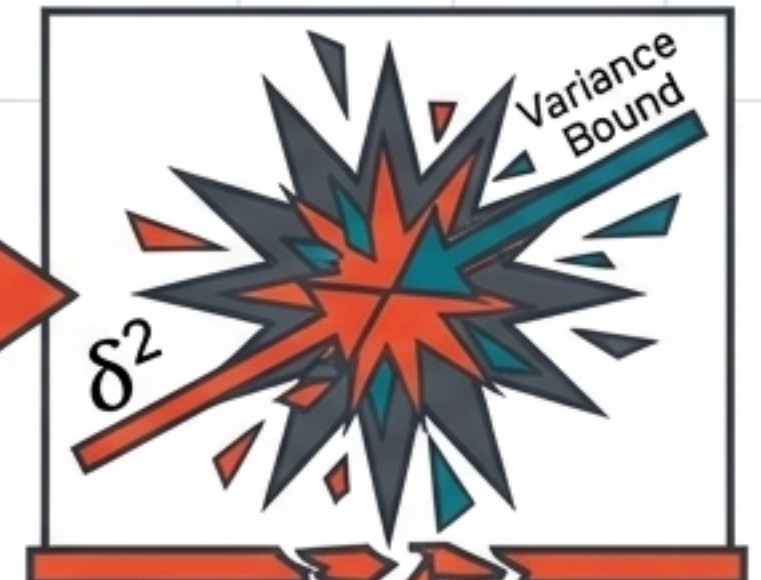
Map: Colorize intervals using displacement map f .



Transport: Use F to rearrange separated blocks.



Average: Average over a hypothetical Følner set.
Variance Bound:
 $\leq 4/D + 4\eta$

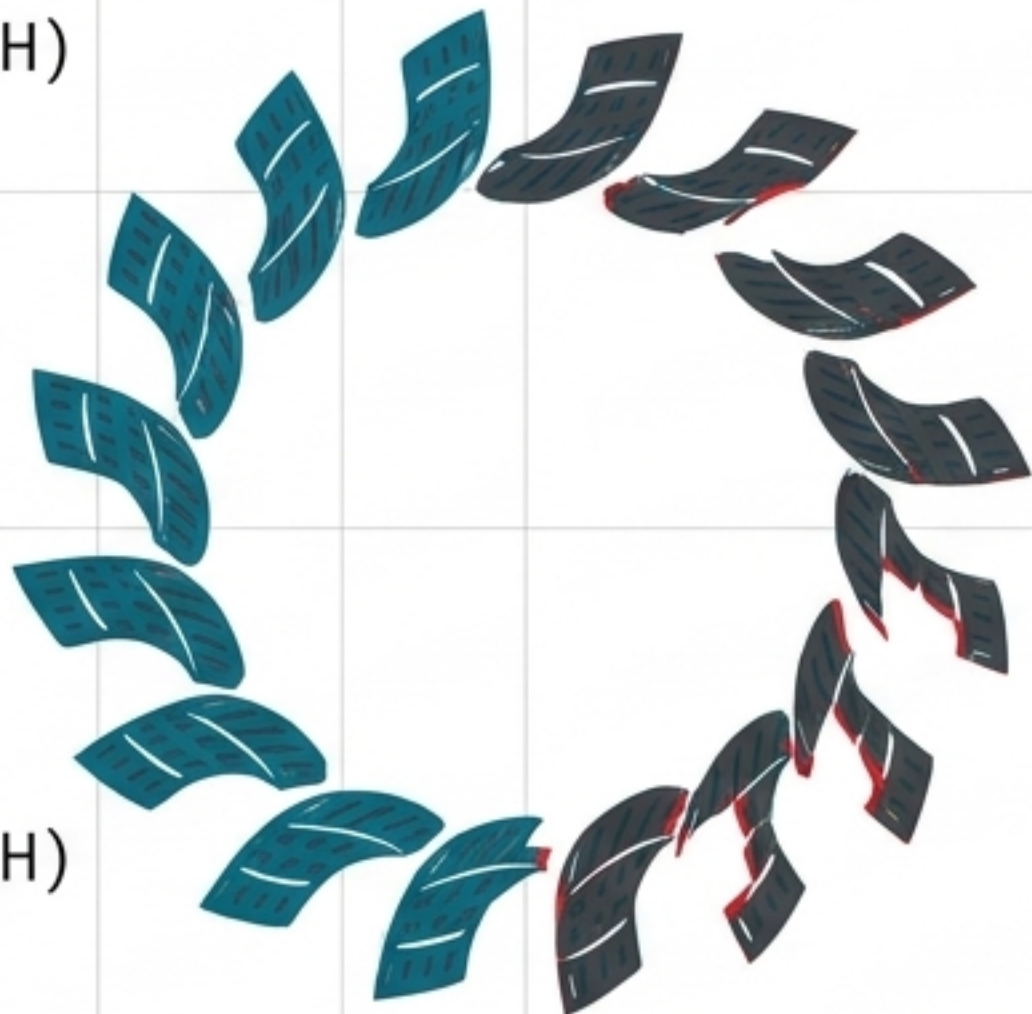


The Snap: Displacement bound (δ^2) collides with Variance Bound.

Domino effects in functional analysis and probability.

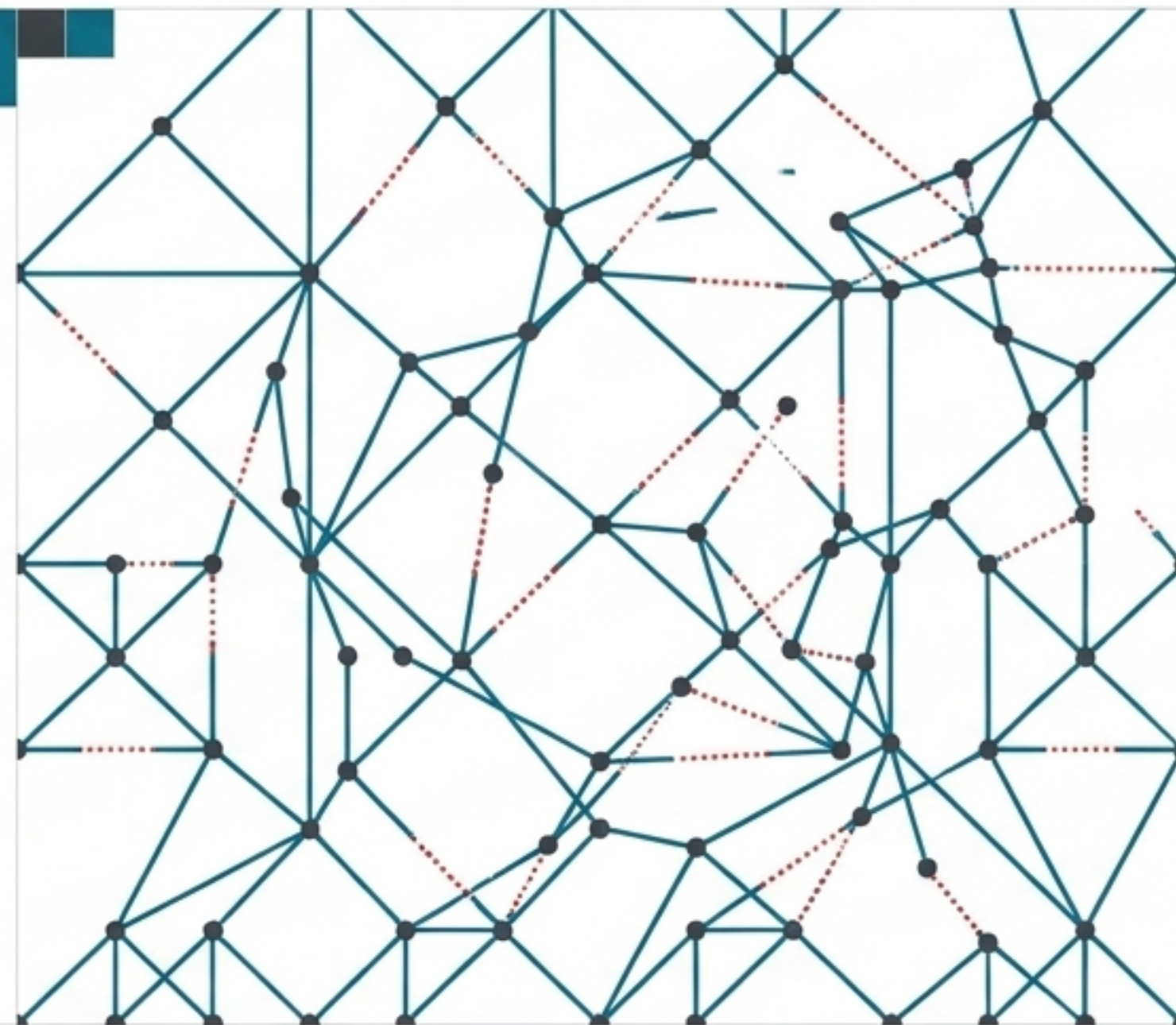
$GL(H)$

$GL(H)$



Representations

Proves there are uniformly bounded representations of $\mathbb{F}$ that cannot be made unitary by any bounded invertible operator.



Probability

Establishes nonuniqueness of percolation on the Cayley graphs of $\mathbb{F}$, guaranteeing infinitely many infinite open clusters between phase thresholds.

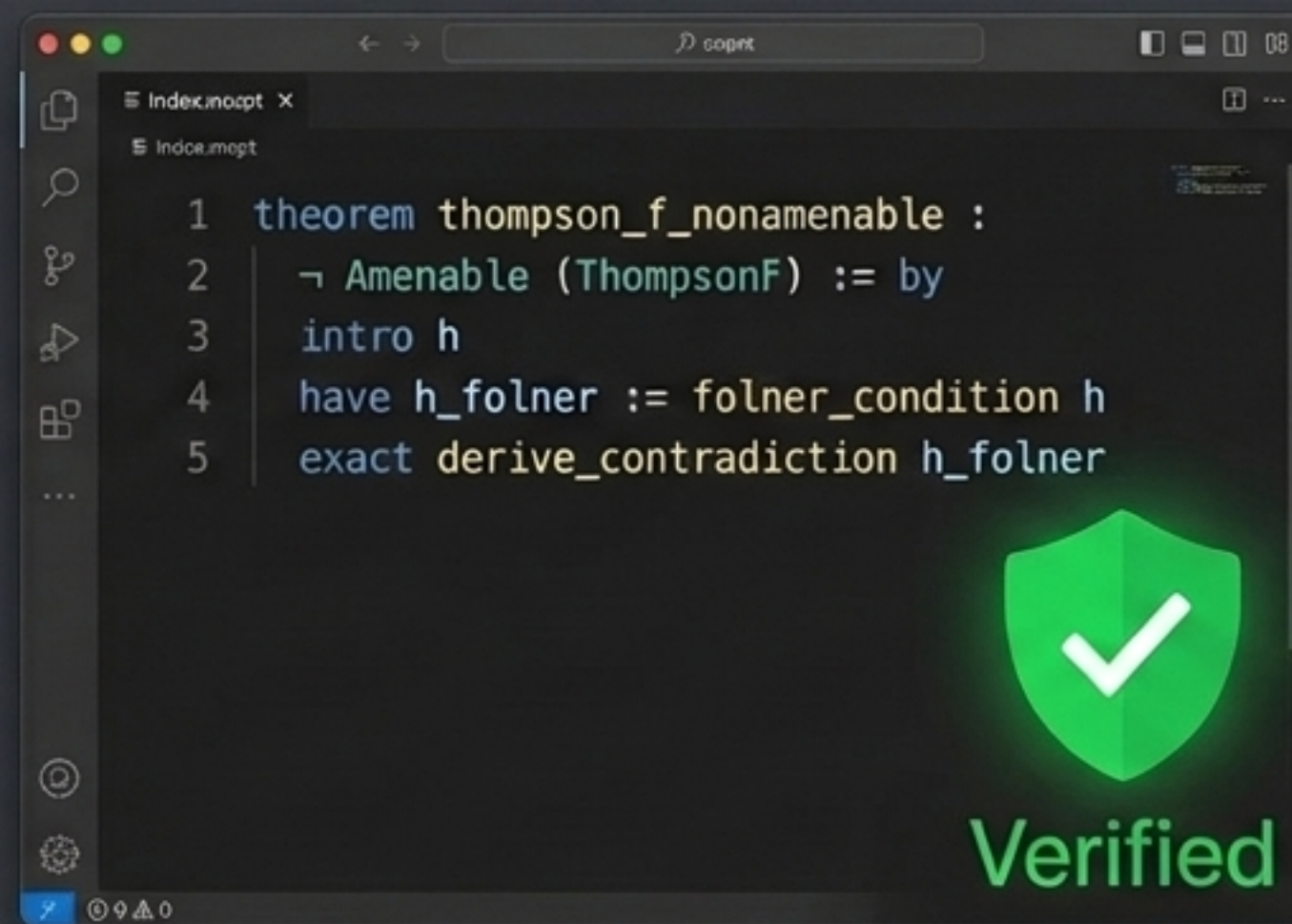
An AI-written proof with a Lean 4 formalization.

The paper was produced by an internal OpenAI model.

Human

The Lean statement is
thompson_F_nonamenable_composition:
composition makes $\mathbb{F}$ a group and
InvariantMean $\mathbb{F}$ is empty.
openai/math lists it as formalized
and checkable with Comparator;
the catalogue review status is
unchecked.

Machine

A screenshot of a code editor window titled 'Index.mocpt x' showing a Lean 4 theorem. The code is as follows:

```
1 theorem thompson_f_nonamenable :  
2   ¬ Amenable (ThompsonF) := by  
3   intro h  
4   have h_folner := folner_condition h  
5   exact derive_contradiction h_folner
```

Below the code, there is a large green shield icon with a white checkmark and the word 'Verified' in green text.



The anomaly is resolved.

Thompson's group F stands as a fully verified, nonamenable entity—proving once and for all that the free group F_2 is not the sole architect of mathematical paradoxes.